

SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 02 : Basic Concepts of Probability and Statistics

LECTURE W2L1 : Introduction to Random Variables and Probability Concepts

BY

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CONCEPTS COVERED

Concepts of Random Experiments, Random Variables and Basic Concepts of Probability

Random Experiments, Sample Space and Events

- **Experiment:** A set of conditions under which the behavior of variables is observed.
- **Random experiment:** An experiment performed under certain conditions where the outcome cannot be predicted with certainty. Each run of a random experiment is generally referred as a trial. Possible outcome(s) of each trial varies (vary); reason to call it random.

  Examples: the number of rainy days in a particular month, measuring the rainfall depth, amount of soil moisture in a soil sample etc.

- **Sample Space:** The set of all possible outcomes of a random experiment is the sample space. Each outcome is called a sample point.

Random Experiments, Sample Space and Events

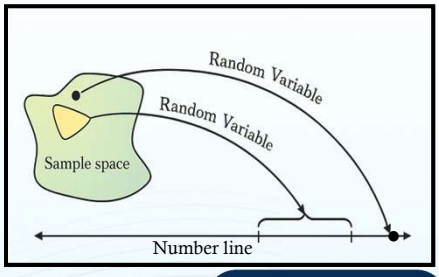
- Sample space example: For the random experiment, 'counting the number of rainy days in June', the sample space consists of only integers from 0 to 30. Outcome of 'measuring the rainfall depth', 'soil moisture' at a location may take any non-negative values.
- The sample space can be classified either as discrete or continuous sample space.
- A sample space is discrete if it has finite or countably infinite elements.
  - For example, the sample space of the random experiment, 'the number of rainy days in April', consists of discrete numbers only (0 to 30).

Random Experiments, Sample Space and Events

- On the other hand, the sample space that consists of a continuum, i.e., all possible values within a range of real numbers, is known as a continuous sample space.
- The sample spaces of 'measuring the rainfall depth', 'soil moisture content' or 'wind speed' at a location are the examples of continuous sample space that consist of any real number in the range of 0 to ∞.
- Event: An event can be defined as a subset of a sample space. Number of rainy days in June equal to 10 is an event from the sample space of 'counting the number of rainy days in June'. Soil moisture below 20% is an event.

Concept of Random Variables and Events

- Random Variable: A Random Variable (RV) is not a variable, rather a function that maps each outcome of an experiment over a sample space to a numerical value on the real line.



- Depending on the nature of the associated sample space Random variable may be discrete or continuous.
- Discrete Random Variable: A random variable that assumes finite or countably infinite values.

Concept of Random Variables and Events

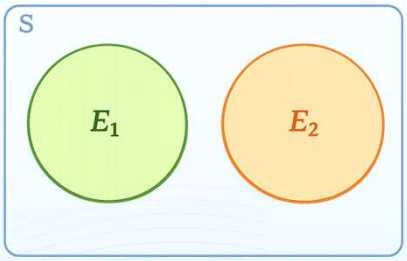
- Continuous Random Variable: A random variable that can take any value within a continuous range of real numbers.
- An example of a discrete random variable would be the 'number of rainy days in June' at a particular location, whereas 'the rainfall depth' at a location is a continuous random variable.
- Any function of a random variable is also a random variable. If X is a random variable then $Z = g(X)$ is also a random variable.

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Concept of Random Variables and Events

Mutually Exclusive Events:

Two events E_1 and E_2 are called mutually exclusive when none of the outcomes in E_1 belongs to E_2 or vice-versa. This is $E_1 \cap E_2 = \boldsymbol{\phi}$, where $\boldsymbol{\phi}$ is null set.



Collectively Exhaustive Events:

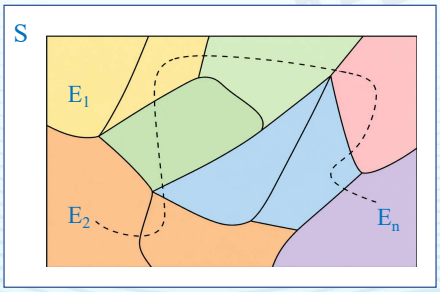
When union of all events ($E_1, E_2 \ldots\ldots, E_n$) comprise the whole sample space, 'S', then $E_1, E_2 \ldots\ldots, E_n$ are called collectively exhaustive events.

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Concept of Random Variables and Events

Mutually Exclusive and Collectively Exhaustive Events:

- Events are called mutually exclusive and collectively exhaustive when they do not overlap with each other, and together they cover the entire sample space.
- This is denoted as: $E_1 \cup E_2 \cup \ldots E_n = S$. Where $E_i \cap E_j = \phi$ for $\forall\ i \neq j$



Basic Concepts of Probability

In any random experiment there is always uncertainty as to whether a specific event will occur or not. The probability concept was proposed originally to explain the uncertainty involved in the outcome of a random experiment. Probability is assigned to the events and this assignment of probability to events is the key for any probabilistic assessment.

- According to the classical definition, the probability of an event A, denoted as $P(A)$, is determined a priori without actual experimentation. It is given by the ratio

$$\mathrm{P}(A) = \frac{N_A}{N}$$

where N is the number of possible outcomes and N_A is the number of outcomes that are favorable to the event A.

Basic Concepts of Probability

Probability is often expressed using relative frequencies.

- If a random event occurs a large number of times N and the event A occurs in n of these occurrences, then the probability of the occurrence of the event A is –

$$\mathrm{P}(A) = \lim_{n\to\infty} \frac{n}{N}$$

The Axioms of Probability

Axioms of Probability: Probability assignment must satisfy certain basic conditions called the axioms of probability.

- **Axiom 1:** The probability of any event lies between 0 and 1:

$$0 \leq P(A) \leq 1$$

- **Axiom 2:** The probability of the entire sample space is 1:

$$P(S) = 1$$

- **Axiom 3:** For mutually exclusive events A and B, the probability of their union equals the sum of their probabilities

$$P(A \cup B) = P(A) + P(\mathrm{B})$$


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 02 : Basic Concepts of Probability and Statistics

LECTURE W2L2 : Some Elementary Properties on Probability

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CONCEPTS COVERED

Some Elementary Properties on Probability

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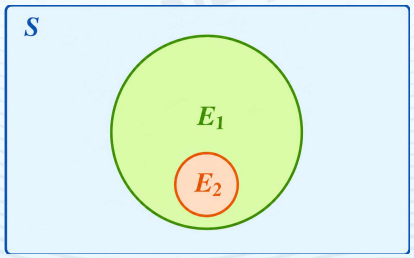
Some Elementary Properties on Probability

- Property 1: If E_1, E_2,..., E_N are mutually exclusive events, then probability

$$P(E_1 \cup E_2 \cup \ldots E_n) = P(E_1) + P(E_2) + \ldots + P(E_n)$$

- Property 2: If event E_2, is a subset of event E_1, then the probability of E_2 is less than or equal to E_1:

$P(E_2) \leq P(E_1)$ and $P(E_1 - E_2) = P(E_1) - P(E_2)$



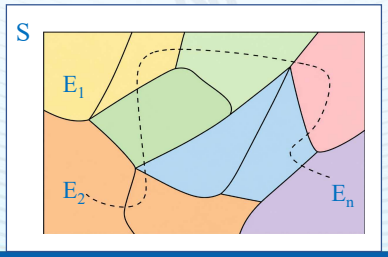
Some Elementary Properties on Probability

- Property 3: If E_1 is complementary to another event E_2 , then

$$P(E_1) = 1 - P(E_2)$$

- Property 4: If an event, E is the union of events E_1, E_2,..., E_n, where E_1, E_2,..., E_n are mutually exclusive,

$$P(E) = P(E_1) + P(E_2) + \ldots + P(E_n)$$

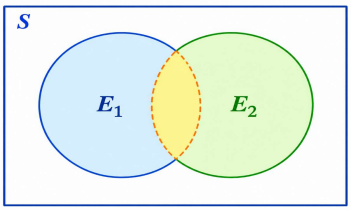


Some Elementary Properties on Probability

- Property 5: For any two events, E_1 and E_2, the probability of E_1 is the sum of the probability of its intersection with E_2 and the complement of E_2

$$P(E_1) = P(E_1 \cap E_2) + P(E_1 \cap E_2^c)$$

- Property 6: For any two events E_1 and E_2, the probability of their union equals the sum of their probabilities minus the probability of their intersection.

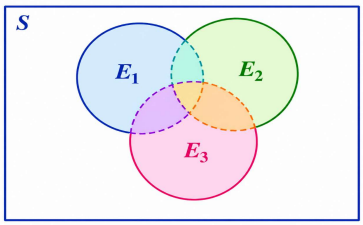


$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

Some Elementary Properties on Probability

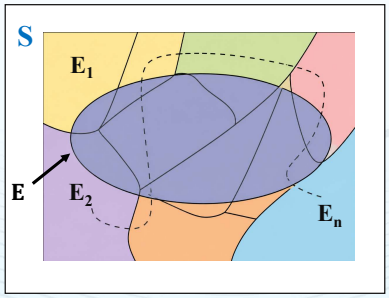
Extending this property, if E_1, E_2 and E_3 are any three events,

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3)$$



Some Elementary Properties on Probability

- Property 7: For mutually exclusive and collectively exhaustive events, E_1, E_2 ……. E_n,



$$P(E) = P(E \cap E_1) + P(E \cap E_2) + \cdots \ldots + P(E \cap E_n)$$

Example

A reservoir system receives water from either a **river**, **tube well**, or **rainfall** source. At any given time, the probability of failure due to inadequate water supply from at least one of these sources is 0.5. Assume that the probabilities of failure of river supply, tube well supply, and rainfall individually are 0.25, 0.10, and 0.30, respectively. Information on their simultaneous failures is given below:

(a) Probability of simultaneous failure of river supply and tube well supply = 0.08

(b) Probability of simultaneous failure of tube well supply and rainfall = 0.04

(c) Probability of simultaneous failure of river supply and rainfall = 0.12

Find the probability of simultaneous failure of all the sources.

Solution

S = failure of river supply; T = failure of tube well supply; R = failure of rainfall

$P(S \cup T \cup R) = 0.5, P(S) = 0.25, P(T) = 0.10, P(R) = 0.30$

$P(S \cap T) = 0.08, P(T \cap R) = 0.04, P(R \cap S) = 0.12$

Probability of simultaneous failure, $P(S \cap T \cap R)$

$P(S \cup T \cup R) = P(S) + P(T) + P(R) - P(S \cap T) - P(T \cap R) - P(R \cap S) + P(S \cap T \cap R)$

$0.5 = 0.25 + 0.10 + 0.30 - 0.08 - 0.04 - 0.12 + P(S \cap T \cap R)$

Thus, $P(S \cap T \cap R) = 0.09$





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Statistical and Machine Learning Methods: Basics and Applications

MODULE 02 : Basic Concepts of Probability and Statistics

LECTURE W2L3 : Descriptive Statistics

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CONCEPTS COVERED

Concept of Descriptive Statistics

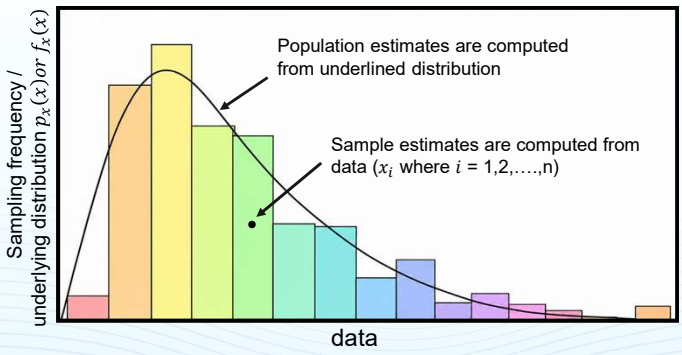
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Descriptive Statistics

- **Probability distributions** describe the probabilistic behavior of random variables when the distribution form and parameters are known.
- **Descriptive statistics** such as central tendency, dispersion, skewness, and tailedness are used to approximately describe population characteristics when the exact distribution is unknown.
- Respective population parameters are the properties of the underlying probability distribution.
- Sample estimates and population parameters are presented simultaneously for better understanding and comparison.

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Descriptive Statistics



Frequency plot of a data set with the underlying distribution used to evaluate the sample estimates (from the data) and population parameters (from the underlying distribution)

Measures of Central Tendency

- The measure of central tendency of a random variable can be expressed in terms of three quantities, namely
  - Mean
  - Median
  - Mode
- The mean can be expressed in different forms such as,
  - Arithmetic mean
  - Geometric mean
  - Weighted mean.

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Measures of Central Tendency

Arithmetic Mean:

- It can be defined as the sum of the observations divided by sample size.
- Consider a sample dataset with n observations $x_1, x_2, \ldots, x_n$ for a random variable X.
- The sample estimate of the population mean (μ) is the arithmetic average $\bar{x}$, calculated as:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

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Measures of Central Tendency

Arithmetic Mean:

- In the case of grouped data,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{k} x_i n_i$$

where k is the number of group, n as the total number of observations, n_i as the number of observations in the ith group and x_i as class mark of the ith group.

- Class mark is defined as midpoint of the group, i.e., mean of upper and lower bounds of the group.

Measures of Central Tendency

Arithmetic Mean:

- For population, considering $p_x(x_i)$ as the underlying distribution (pmf) of a discrete random variable X, the population mean μ is expressed as,

$$\mu = \sum_{i=1}^{n} x_i p_X(x_i)$$

- Considering $f_X(x)$ as the underlying distribution (pdf) of a continuous random variable X, the population mean μ is expressed as

$$\mu = \int_{-\infty}^{\infty} x\, f_X(x)\, dx$$

Measures of Central Tendency

Geometric Mean:

- The geometric mean indicates the central tendency of a data set by using the product of their values.
- The geometric mean ($\overline{x_G}$) is the nth root of the product of n observations. $\overline{x_G}$ can be evaluated:

$$\overline{x_G} = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}$$

where the symbol Π is implies multiplication.

- The geometric mean of the population is expressed as:

$$\mu_G = \text{antilog}\,[\,E\,(log\,X)]$$

where E stands for expectation.

Measures of Central Tendency

Weighted Mean:

The weighted mean is similar to an arithmetic mean except some data points contribute more than others. The weighted mean is:

$$\overline{x_w} = \frac{\sum_{i=1}^{k} x_i w_i}{\sum_{i=1}^{k} w_i}$$

where w_i is the weighted associated with the ith observation or group and k is the number of observations or groups.

Measures of Central Tendency

Median:

- The median is the value of the random variable at which the values on both sides of it are equally probable.
- For odd number of observations (n), the median is the $(n+1)/2^{th}$ observation after arranging the data in ascending order.
- For even number of observations (n), the median is the average of the $(n/2)^{th}$ and $(n/2+1)^{th}$ observations.
- The median (x_{md}) divides the data into two equal halves, such that half the observations lie below it and half lie above it.

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Measures of Central Tendency

Median:

- For a discrete random variable, the population median $\mu_{md} = x_d$, where d is determined from

$$\sum_{i=1}^{d} p_x(x_i) = 0.5$$

- For a continuous random variable, the population median μ_{md} is calculated from

$$\int_{-\infty}^{\mu_{md}} f_x(x)dx = 0.5$$

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Measures of Central Tendency

Mode:

- The mode is the most probable or most frequently occurring value of a random variable.
- It is the value of the random variable with the highest probability density or the most frequently occurring value.
- A sample or a population may have none, one or more than one mode.
- For a discrete random variable with pmf $p_x(x)$, the mode is the value of x_i for which $p_x(x_i)$ is maximum, i.e.,

$$\mu_{mo} = \arg max_{x_i} [p_x(x_i)]$$

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Measures of Central Tendency

- For a discrete random variable with *pmf* $p_x(x)$, the mode is the value of x_i for which $p_x(x_i)$ is maximum, i.e.,

$$\mu_{mo} = \arg max_{x_i} [p_x(x_i)]$$

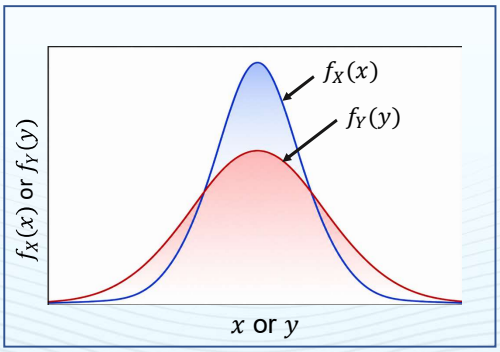
- For a continuous random variable, X with *pdf* $f_x(x)$, the mode is the value of X,

$$\frac{df_x(x)}{dx} = 0 \quad \text{and} \quad \frac{d^2 f_x(x)}{dx^2} < 0$$

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Measure of Dispersion

The dispersion of a random variable corresponds to how closely the values of a random variable are clustered or how widely it is spread around the central value.



Random variables X and Y with the same mean but different dispersion

Measure of Dispersion

Range: The range of a sample is the difference between the maximum and the minimum values in the sample.

Variance: Variance is a measure of the dispersion of a random variable taking the mean as the central value.

- For a sample of size n, the variance is the average squared deviation from the sample mean.
- The sample estimate of variance can be defined as the average of the squared deviations from the mean.

Measure of Dispersion

- The sample estimate of population variance σ^2 is denoted by S^2, and is given as:

$$S^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

- A random variable X and a sample $x_1, x_2, \ldots\ldots x_n$ with sample mean $\bar{x}$.
- The reason for dividing by $n-1$ instead of n is to make the estimator unbiased.
- For the grouped data with $x_1, x_2, \ldots\ldots x_k$ as the class mark, the variance can be:

$$S^2 = \frac{\sum_{i=1}^{k}(x_i - \bar{x})^2 n_i}{n-1}$$

where k is the number of groups, n is the total number of observations.

Measure of Dispersion

- **Standard deviation (S)**, another measure of dispersion, is the positive square root of variance.
- The formula of standard deviation S:

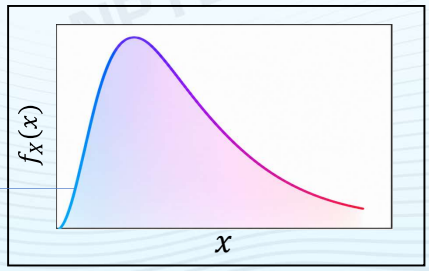
$$S = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

- **Coefficient of variation (C_v)** is defined as the standard deviation divided by the mean.

$$C_v = \frac{S}{\bar{x}}$$

Measure of Symmetry

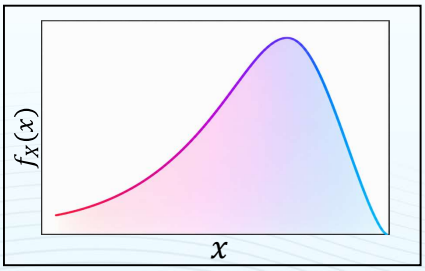
- Data distributions may not always be symmetric about the mean and can extend more toward the right or left side. Such distributions are said to be skewed.
- Skewness of the data is measured using the coefficient of skewness (γ).
- When the data is skewed to right side and $\gamma > 0$, called positive skewness.



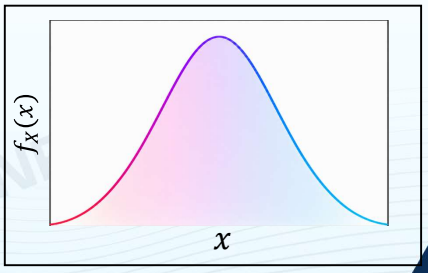
Positively skewed

Measure of Symmetry

- When the distribution is skewed to left side and $\gamma < 0$, called negative skewness and whereas $\gamma = 0$ represents symmetric distribution.



The pdf plot of a negatively skewed distribution



The pdf plot of a symmetric distribution

Measure of Symmetry

- The difference between the mean and the mode indicates the skewness of the data.
- The sample estimate skewness is normally made dimensionless by dividing by S^3 to get the coefficient of skewness.
- The coefficient of skewness is expressed :

$$C_S = \frac{n\sum_{i=1}^{n}(x_i - \bar{x})^3}{(n-1)(n-2)S^3}$$

Measure of Tailedness

- The measure of tailedness of a probability distribution function is referred to as kurtosis.
- For a sample, kurtosis shows the effect of existing outliers.
- The kurtosis is made dimensionless by dividing by S^4 to get the coefficient of kurtosis.
- The coefficient of kurtosis of sample estimate is given by :

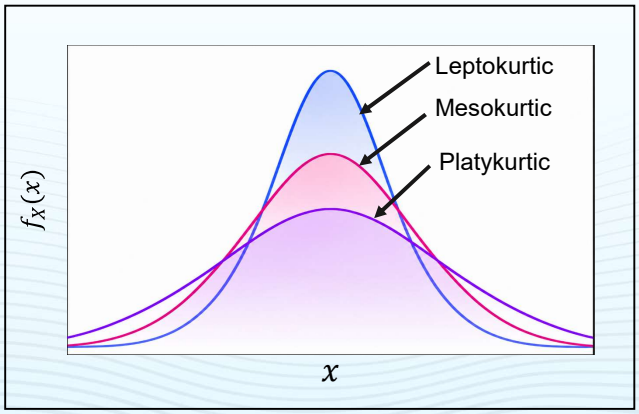
$$k = \frac{n^2\sum_{i=1}^{n}(x_i - \bar{x})^4}{(n-1)(n-2)(n-3)S^4}$$

Measure of Tailedness

- Generally, the standard value taken is the kurtosis of normal distribution that has a value of 3 .
- Based on the measure of kurtosis, data or the associated distribution can be divided into three types as follows :
- Mesokurtic: If any distribution has the same kurtosis as compared to normal distribution, the distribution is called Mesokurtic and $k = 3$.
- Leptokurtic: In case a distribution has a relatively greater concentration of probability near the mean than the normal distribution and $k > 3$.

Measure of Tailedness

- Platykurtic: In case a distribution has a relatively smaller concentration of probability near the mean than the normal distribution and $k < 3$.



Example

Consider the following sample data for annual maximum rainfall (mm) at a meteorological station. Evaluate the mean, variance, standard deviation coefficient of skewness and coefficient of kurtosis.

Year	Rainfall (mm)	Year	Rainfall (mm)
2001	820	2009	1060
2002	950	2010	875
2003	760	2011	910
2004	1100	2012	1340
2005	980	2013	780
2006	890	2014	990
2007	1205	2015	1150
2008	730	2016	860

Solution:

1. Mean

Mean is calculated as:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\bar{x} = \frac{820 + 950 + \cdots + 860}{16}$$

$$\bar{x} = \frac{15360}{16}$$

$$\bar{x} = 960\ mm$$

Solution:

2. Variance

$$S^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

$$S^2 = \frac{(820-960)^2+(950-960)^2+\ldots.+(860-960)^2}{16-1}$$

$$S^2 = 28976.67.67$$

3. Standard Deviation

Standard deviation$= \sqrt{S^2}$

Standard deviation$= \sqrt{28976.67} = 170.225$

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Solution:

4. Coefficient of Skewness:

$$C_S = \frac{n\sum_{i=1}^{n}(x_i - \bar{x})^3}{(n-1)(n-2)S^3}$$

$$C_S = \frac{16 * [(820-960)^3 + (950-960)^3 + \cdots + (820-960)^3]}{(16-1)(16-2) * 170.225^3}$$

$$C_S = 0.75$$

5. Coefficient of kurtosis:

$$k = \frac{n^2\sum_{i=1}^{n}(x_i - \bar{x})^4}{(n-1)(n-2)(n-3)S^4}$$

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Solution:

$$k = \frac{16^2 * [(820 - 960)^4 + (950 - 960)^4 + \cdots + (820 - 960)^4]}{(16 - 1) \times (16 - 2) \times (16 - 3) \times 170.225^3}$$

$$k = 3.592$$

Therefore, $k > 3$, so the distribution is Leptokurtic.





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MODULE 02 : Basic Concepts of Probability and Statistics

LECTURE W2L4: Total Probability Theorem and Bayes Rule

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CONCEPTS COVERED

Conditional Probability, Total Probability Theorem, and Bayes Rule

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Conditional Probability Theorem

Definition:

- The conditional probability theorem describes the probability of an event occurring when another event has already occurred.
- If A and B are two events in a sample space S, and $P(A) \neq 0$, the conditional probability of event B given that A has already occurred is

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \Rightarrow P(A \cap B) = P(A)\,P(B|A)$$

- Similarly, given that $P(B) \neq 0$.

$$P(A|B) = \frac{P(B \cap A)}{P(B)} \Rightarrow P(A \cap B) = P(B)\,P(A|B)$$

Conditional Probability Theorem for multiple events

- Let the E_1, E_2, and E_3 be three events.
- For all three events, the probability of E_1 times the E_2 given that E_1 has occurred, times the probability of E_3 given that both E_1 and E_2 have already occurred.

$$P(E_1 \cap E_2 \cap E_3) = P(E_1)\,P(E_2|E_1)\,P(E_3|E_1 \cap E_2)$$

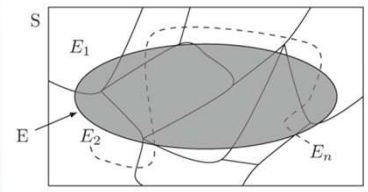
- The theorem can be applied to n events

$$P(E_1 \cap E_2 \cap \cdots \cap E_n)$$
$$= P(E_1)\,P(E_2|E_1)\,P(E_3|E_1 \cap E_2)\ldots\,P(E_n|E_{n-1} \cap \cdots \cap E_2 \cap E_1)$$

Total Probability Theorem

- Let $E_1, E_2, \ldots E_n$ be the n events, which are mutually exclusive and collectively exhaustive events.
- The event A also belongs to the same sample space
- The $P(A)$ can be evaluated through E_n events,

$$P(A) = P(A \cap E_1) + P(A \cap E_2) + \ldots + P(A \cap E_n)$$



- From the conditional probability we can write:

$$P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + \ldots + P(E_n)P(A|E_n)$$

- Here, $P(E_1), P(E_2), \ldots$ etc. are weights with $\sum_{i=1}^{n} P(E_i) = 1$
- Therefore, $P(A) = \sum_{i=1}^{n} P(E_i)P(A|E_i)$,

this is the Total Probability Theorem

Bayes' Rule

- From the Total Probability Theorem, we are interested in finding any probability of $P(E_i)$, given that the event A has already occurred.
- The conditional probability theorem can be used as follows:

$$P(A \cap E_i) = P(E_i \cap A)$$

- That leads to $P(E_i)P(A|E_i) = P(A)P(E_i|A)$
- On re-arranging, $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{P(A)}$
- Utilizing the Total Probability Theorem from the expression, we get:

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{i=1}^{n} P(E_i)P(A|E_i)}$$

this is known as Bayes' rule

Bayes Rule

- The advantage of Bayes theorem is that the posterior probability is now expressed in terms of quantities that can be estimated.

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{i=1}^{n} P(E_i)P(A|E_i)}$$

- The denominator of the Bayes theorem is a constant term, i.e. the total probability $P(A)$ that is independent of the events, E_i.
- Thus, denominator term may be taken out, and using proportionality, we get

$$P(E_i|A) \propto P(E_i)P(A|E_i)$$

Bayes Rule

- Now, in this expression:

$$P(E_i|A) \propto P(E_i)P(A|E_i)$$

- The term $P(E_i)$ is the probability of occurrence of E_i, without knowing any other information. This term is referred as *prior*.
- Next, knowing that event A has occurred, probability of occurrence of E_i, $P(E_i|A)$ is updated. Thus, this term is referred as *posterior*.
- The probability of occurrence of the event A, given that E_i has occurred, is generally evaluated/estimated from historical records or experience. This term $P(A|E_i)$ is referred as *likelihood*.
- Using these terms, Bayes' rule is often expressed as

$$Posterior \propto Prior \times Likelihood$$

Example

A series of rainfall record is assimilated from the measurements obtained from 3 different instruments. 25% measurements are taken by instrument A that yields one missing data out of 200 on an average, 50% measurements are taken by instrument B that yields one missing data out of 150 on an average and rest by instrument C that yields one missing data out of 100 on an average. One measurement is found to yield a missing data, what is the probability that the measurement was taken by instrument A.

Solution

The probability that the measurement was made by instrument A on condition that the measurement is wrong can be calculated using Bayes' Theorem.

Let X_1,X_2, and X_3 represent the events that measurement was made by instrument A, B and C respectively.

Solution

Let Y represents the event that the measurement was missing.

$$P(X_1|Y) = \frac{P(Y|X_1)P(X_1)}{\sum_{i=1}^{3} P(Y|X_i)P(X_i)}$$

Hence,

$$P(X_1|Y) = \frac{\left(\frac{1}{200}\right) \times 0.25}{\left(\frac{1}{200}\right) \times 0.25 + \left(\frac{1}{150}\right) \times 0.5 + \left(\frac{1}{100}\right) \times 0.25} = 0.176$$

Therefore, the probability that the measurement was made by instrument A, given that the measurement is wrong is **0.176**.

Example

A series of soil moisture data is prepared by collecting samples from two different sources. Though the sources are random for any month, a total of 550 samples are obtained from source – A that contains 3% erroneous data and a total of 350 samples are obtained from source – B that contains 1% erroneous data.

a. What is the probability that the data for a month selected at random is obtained from source – A?

b. What is the overall percentage of erroneous data?

c. An erroneous data is selected at random, what is the probability that it is from source – A?

Solution

Let us denote the following events:

A: data obtained from source – A

B: data obtained from source – B

E: selected data is erroneous

a. The probability of data obtained from source – A, i.e., $P(A)$ is given by

$$P(A) = \frac{550}{550 + 350} = 0.611$$

b. The erroneous data may come from either source – A or source – B. Therefore, we need to apply that total probability theorem to calculate the probability of event E.

Solution

b. Therefore, the probability of Event E, i.e., the selected value is erroneous:

$$P(E) = P(E|A)P(A) + P(E|B)P(B)$$

$$P(E) = 0.03 \times 0.611 + 0.01 \times 0.389$$

$$P(E) = 0.022$$

c. If the sample data selected at random is erroneous, probability that it comes from source – A is not 0.611 as in case of solution (a), it is because of the change of sample space. Instead of entire data, the new sample space consists of only erroneous data.

Solution

c. Thus, using Bayes' rule,

$$P(A|E) = \frac{P(E|A)P(A)}{P(E|A)P(A) + P(E|B)P(B)}$$

$$P(A|E) = \frac{0.03 \times 0.611}{0.03 \times 0.611 + 0.01 \times 0.389} = 0.825$$

Therefore, the probability that the selected erroneous data is from source-A is **0.825**.


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ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 02 : Basic Concepts of Probability and Statistics

LECTURE W2L5 : Moments and Expectation

BY

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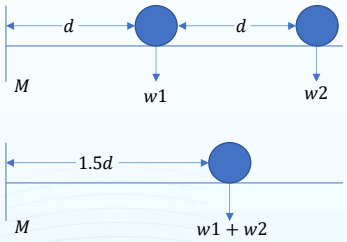
CONCEPTS COVERED

Concept of Moments and Expectation

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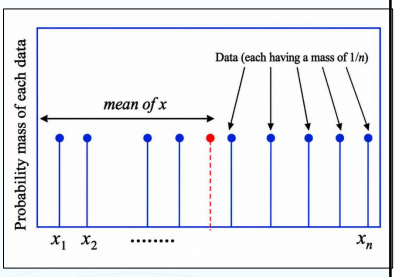
Concept of Moments

- In physics, moment is the product of a physical quantity and the distance from a fixed point of reference.
- While considering mass as a physical quantity, it can be used to locate the center of gravity of any irregularly shaped object, and higher-order of moments can also be evaluated.
- Similar concepts can be utilized to extract some meaningful information from a dataset.



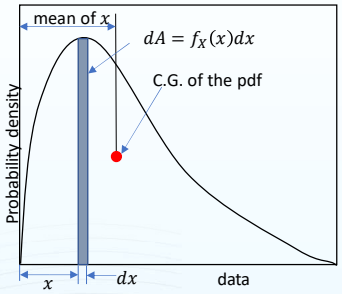
Concept of Moments

- Suppose that the data $x_1, x_2, \ldots, x_n$ is located according to their values on the real line.
- Assuming that each data value is equiprobable, the mass of each data point can be assumed to be $1/n$, when n is the total number of the data.
- Now, the total moment with respect to the origin can be evaluated as $\sum_{all\ i} x_i\ (1/n)$.
- The locations say $\tilde{x}$ of the equivalent total mass, i.e., the mass that will create the same moment (as the total moment) about the origin, expressed as $(n \times 1/n)\bar{x} = \bar{x}$.
- On equating these two moments, we get $\bar{x} = \sum_{all\ i} x_i(\frac{1}{n})$.
- This location $(\bar{x})$ is equivalent to the mean of the data.



Concept of Moments

- In case of the population, which is represented by a pdf, the mean can be identified following the same concept.
- Consider a delta width (dx) located at a distance x at a distance x from the origin.
- The total probability mass is equal to the shaded area in the figure $= dA$.
- The total moment for this area with respect to origin is $x.dA = x.f_X(x)dx$.
- On integrating the entire range of the data $(-\infty, \infty)$, the total moment can be written as $\int_{-\infty}^{+\infty} xf_X(x)dx$.



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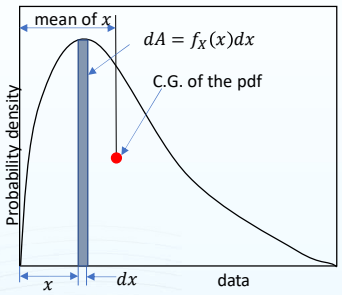
Concept of Moments

- If it is assumed that the total probability mass is located at a distance x from origin that produces same amount of moment, we may write

$$\mu \times \int_{-\infty}^{\infty} f_X(x)\,dx = \int_{-\infty}^{+\infty} xf_X(x)dx$$

Since $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$

$$\mu = \int_{-\infty}^{+\infty} xf_X(x)dx$$



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Higher order Moments

- Following the same concept, higher-order moments with respect to the origin can also be evaluated using some power of distance from the origin.
- The x can be replaced with x^i to evaluate the ith moment with respect to the origin.
- In probability theory, the first moment is calculated with respect to origin, which is mean.
- However, the second moment onwards, these are calculated with respect to the mean.

Second order Moments

- The second-order moment with respect to the mean can be evaluated as

$$E[(X-\mu)^2] = \int_{-\infty}^{+\infty} (x-\mu)^2 f_X(x)dx$$

- In general, the ith order moment with respect to the mean can be evaluated as

$$E\left[(X-\mu)^i\right] = \int_{-\infty}^{+\infty} (x-\mu)^i f_X(x)dx$$

Expectation

The **expected value** of a random quantity intuitively means the **averaged value** of the outcome of the corresponding random experiment carried out repetitively for an infinite number of times.

- Mathematically, the expected value of a random variable (X), represented as $E(X)$, can be defined as the first moment about the origin and represented as follows:

$$E(X) = \mu$$

- Considering X to be a discrete random variable, the expected value of X is given as

$$E(X) = \sum_{al\ j} x_j p_X(x_j)$$

Expectation

- For continuous random variables, the expected value of X is given as

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Any function X, say $g(X)$ is also a random variable. Thus, the expected value of $g(X)$ is given as

$$E[g(X)] = \sum_{al\ j} g(x_j) p_x(x_j) \quad \text{for discrete RV}$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx \quad \text{for continuous RV}$$

Expectation

Relating the concept of moment with expectation, the following points can be noted

i. The **first moment** about the **origin** is the ***mean***, i.e.,

$$E(X) = \mu = \begin{cases} \sum_{all\ j} x_j p_X x_j & \text{for discrete RV} \\ \int_{-\infty}^{\infty} x f_X(x) dx & \text{for continuous RV} \end{cases}$$

Expectation

Some useful information on the *expected values* are:

i. Expectation of a constant is the same as that constant, i.e.,$E(C) = C$.

ii. Expectation of a modified random variable obtained by multiplying with a constant is equal to the product of the constant and the expectation of the original random variable, i.e., $E(CX) = CE(X)$.

iii. Expectation of a random variable obtained by addition/subtraction of two random variables is equal to the sum/difference of their individual expectations, i.e., $E(X \pm Y) = E(X) \pm E(Y)$.

Second moment using Expectation

ii. The **second moment** about the mean is the ***variance***, i.e.,

$$E(X-\mu)^2 = \sigma^2 = Var(x) = \begin{cases} \sum_{al\ j} (x_j - \mu)^2 p_X(x_j) & \text{for discrete RV} \\ \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x)dx & \text{for continuous RV} \end{cases}$$

It can also be shown that

$$Var(x) = E(x^2) - [E(x)]^2$$

Expectation

Some useful information on the *variance values* are:

i. Variance of a constant is zero, i.e., $Var(C) = 0$.

ii. Variance of a modified random variable obtained by multiplying with a constant is equal to the product of the square of constant and the variance of the original random variable, i.e., $Var(CX) = C^2 Var(X)$.

iii. Variance of a modified random variable obtained by multiplying by a constant (a) and adding to another constant, (b) is equal to the product of the square of constant (a) and the variance of the original random variable, i.e.,

$$Var(aX + b) = a^2 Var(X)$$

Third moment using Expectation

iii. The **third moment** about the **mean** is the ***skewness***

$$E[(X-\mu)^3] = \begin{cases} \sum_{all\ j} (x_j-\mu)^3 p_X(x_j) & \text{for discrete RV} \\ \int_{-\infty}^{\infty} (x-\mu)^3 f_X(x)dx & \text{for continous RV} \end{cases}$$

It can also be shown that

$$E[(X-\mu)^3] = E(x^3) - 3E(x^2)E(x) + 2\{E(x)\}^3$$

The measure of skewness is non-dimensionalized using the variance and is termed the *coefficient of skewness* (γ).

$$\gamma = \frac{E[(X-\mu)^3]}{\sigma^3}$$

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Fourth moment using Expectation

iv. The **fourth moment** about the **mean** is the ***kurtosis*** *(measure of tailedness)*,

$$E[(X-\mu)^4] = \begin{cases} \sum_{all\ j} (x_j-\mu)^4 p_X(x_j) & for\ discrete\ RV \\ \int_{-\infty}^{\infty} (x-\mu)^4 f_X(x)dx & for\ continous\ RV \end{cases}$$

It can also be shown that

$$E[(X-\mu)^4] = E(x^4) - 4E(x^3)E(x) + 6E(x^2)\big(E(x)\big)^2 - 3\{E(x)\}^4$$

The measure of tailedness *(kurtosis)* is non-dimensionalized using the variance and is termed as *coefficient of kurtosis* (κ).

$$\kappa = \frac{E[(X-\mu)^4]}{\sigma^4}$$

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Example

The number of thunderstorms per year during a given year is represented by the random variable X. The pmf obtained from the historical data are shown the following table:

X	0	1	2	3	4	5	6
$pmf\ (p_X(x))$	0.10	0.20	0.25	0.20	0.15	0.07	0.03

What is the mean and variance of the number of thunderstorms in a year?

Solution

The mean of number of thunderstorms per year can be evaluated as

$$E(X) = \sum xp_X(x)$$

$$= 0(0.1) + 1(0.2) + 2(0.25) + 3(0.2) + 4(0.15) + 5(0.07) + 6(0.03)$$

$$= \mathbf{2.43}$$

Thus, the mean of no. of thunderstorms in a year is **2.43**.

The variance of storms can be evaluated as

$$Var(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \sum x^2 p_X(x)$$

Solution

$$E(X^2) = 0^2(0.1) + 1^2(0.2) + 2^2(0.25) + 3^2(0.2) + 4^2(0.15) + 5^2(0.07) + 6^2(0.03)$$
$$= \mathbf{8.23}$$
$$Var(X) = E(X^2) - [E(X)]^2$$
$$= 8.23 - (2.43)^2$$
$$= \mathbf{2.325}$$

Thus, the variance of no. of thunderstorms in a year is **2.325**

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Example

The daily discharge X (in m^3/s) observed at a river gauging station modeled using the following probability density function:

$$f_X(x) = \begin{cases} \frac{x}{98}, & 0 \leq x < 14 \\ 0, & \text{otherwise} \end{cases}$$

Calculate

a) the expected value of daily discharge, i.e. $E(X)$

b) the variance of the daily discharge, i.e. $Var(X)$

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Solution

Given pdf of daily discharge X (in m^3/s);

$$f(x) = \begin{cases} \frac{x}{98}, & 0 \leq x < 14 \\ 0, & \text{otherwise} \end{cases}$$

i. The expected daily discharge,

$$E(X) = \int_0^{14} xf(x)dx = \int_0^{14} x\frac{x}{98}dx$$

$$= \frac{1}{98}\left[\frac{x^3}{3}\right]_0^{14} = \frac{1}{98}\frac{2744}{3} = \mathbf{9.334}\ m^3/s$$

Solution

Variance of daily discharge,

$$V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_0^{14} x^2 f(x)dx$$

$$E(X^2) = \int_0^{14} x^2\frac{x}{98}dx = \frac{1}{98}\left[\frac{x^4}{4}\right]_0^{14} = \frac{1}{98}\frac{38416}{4} = \mathbf{98}$$

$$V(X) = 98 - (9.334)^2$$
$$= 98 - 87.11$$
$$= 10.89$$

Thus, the expected daily discharge, $\boldsymbol{E(X) = 9.334}\ m^3/s$

and, the variance of daily discharge $\boldsymbol{Var(X) = 10.89}\ (m^3/s)^2$


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